numpy

```
import numpy as np
```

```
np.__version__
```

```
'2.2.4'
```

```
a = np.arange(1,10)
a
```

```
array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
np.arange(10)
```

```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
np.arange(1,11,2)
```

```
array([1, 3, 5, 7, 9])
```

```
np.arange(10,1,-1)
```

```
array([10,  9,  8,  7,  6,  5,  4,  3,  2])
```

```
a = a.reshape(3,3)
```

```
a
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

```
type(a)
```

```
numpy.ndarray
```

```
a + 1
```

```
array([[ 2,  3,  4],
       [ 5,  6,  7],
       [ 8,  9, 10]])
```

```
a.shape
```

```
(3, 3)
```

```
a.size
```

```
9
```

```
a.ndim
```

```
2
```

```
a.dtype
```

```
dtype('int64')
```

```
a.itemsize
```

```
8
```

```
a = np.array([[1,2,3], [4,5,6]], dtype='int64')
a
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
a.T   # transpose
```

```
array([[1, 4],
       [2, 5],
       [3, 6]])
```

```
a.flatten()
a.ravel()
```

```
array([1, 2, 3, 4, 5, 6])
```

```
a = np.zeros([3,3], dtype=np.int64)
a
```

```
array([[0, 0, 0],
       [0, 0, 0],
       [0, 0, 0]])
```

```
b = np.ones([3,3])
b
```

```
array([[1., 1., 1.],
       [1., 1., 1.],
       [1., 1., 1.]])
```

```
a + b
```

```
array([[1., 1., 1.],
       [1., 1., 1.],
       [1., 1., 1.]])
```

```
np.full([3,3], 7)
```

```
array([[7, 7, 7],
       [7, 7, 7],
       [7, 7, 7]])
```

```
np.linspace(10,20,3)
```

```
array([10., 15., 20.])
```

```
np.logspace(10,20,3)
```

```
array([1.e+10, 1.e+15, 1.e+20])
```

```
a = np.array([[1,2,3], [4,5,6], [7,8,9]])
a
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

```
a.max()
```

```
np.int64(9)
```

```
a.max(axis=0)  # maximun in each Column
```

```
array([7, 8, 9])
```

```
a.max(axis=1)  # maximun in each Row
```

```
array([3, 6, 9])
```

```
a.sum()
```

```
np.int64(45)
```

```
a.sum(axis=0)  # sum in each Column
```

```
array([12, 15, 18])
```

```
a.sum(axis=1)  # sum in each Row
```

```
array([ 6, 15, 24])
```

```
a.cumsum()  # cumulative
```

```
array([ 1,  3,  6, 10, 15, 21, 28, 36, 45])
```

```
a.cumsum(axis=0)
```

```
array([[ 1,  2,  3],
       [ 5,  7,  9],
       [12, 15, 18]])
```

```
a.cumsum(axis=1)
```

```
array([[ 1,  3,  6],
       [ 4,  9, 15],
       [ 7, 15, 24]])
```

```
a = np.array([1,2,3])
b = np.array([4,5,6])
```

```
print(a)
print(b)
```

```
[1 2 3]
[4 5 6]
```

```
print(a.dot(b))  # 1*4 + 2*5 + 3*6
```

```
32
```

```
a * b
```

```
array([ 4, 10, 18])
```

```
a = np.array([[1,4,2], [3,4,6], [0,-1,5]])
a
```

```
array([[ 1,  4,  2],
       [ 3,  4,  6],
       [ 0, -1,  5]])
```

```
np.sort(a, axis=0)
```

```
array([[ 0, -1,  2],
       [ 1,  4,  5],
       [ 3,  4,  6]])
```

```
np.sort(a, axis=1)   # default --> axis=1
```

```
array([[ 1,  2,  4],
       [ 3,  4,  6],
       [-1,  0,  5]])
```

```
np.sort(a, axis=None)
```

```
array([-1,  0,  1,  2,  3,  4,  4,  5,  6])
```

```
print(np.sqrt(4))
```

```
2.0
```

```
print(np.sin(np.pi/2))
```

```
1.0
```

```
a = np.array([1,2,3])
```

```
np.sqrt(a)
```

```
array([1.        , 1.41421356, 1.73205081])
```

```
np.cos(a)
```

```
array([ 0.54030231, -0.41614684, -0.9899925 ])
```

```
a = np.array([4,5,2,1,3,9])
a
```

```
array([4, 5, 2, 1, 3, 9])
```

```
print(a[1])
print(a[[1,2]])
```

```
5
[5 2]
```

```
a = np.array([[1,2,3], [4,5,6]])
a
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
for i in a:
    print(i)
```

```
[1 2 3]
[4 5 6]
```

```
for i in a.flatten():
    print(i)
```

```
1
2
3
4
5
6
```

```
a
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
a.reshape(3,2)  # return a copy
```

```
array([[1, 2],
       [3, 4],
       [5, 6]])
```

```
a.resize(3, 2)  # change the origin
```

```
a
```

```
array([[1, 2],
       [3, 4],
       [5, 6]])
```

```
np.resize(a, [2,3])
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
a
```

```
array([[1, 2],
       [3, 4],
```

```
       [5, 6]])
```

```
a = np.array([[1,2,3], [4,5,6]])
b = np.array([[7,8,9], [10,11,12]])
```

```
a
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
b
```

```
array([[ 7,  8,  9],
       [10, 11, 12]])
```

```
np.hstack([a, b])
```

```
array([[ 1,  2,  3,  7,  8,  9],
       [ 4,  5,  6, 10, 11, 12]])
```

```
np.vstack([a,b])
```

```
array([[ 1,  2,  3],
       [ 4,  5,  6],
       [ 7,  8,  9],
       [10, 11, 12]])
```

```
np.column_stack([a,b])
```

```
array([[ 1,  2,  3,  7,  8,  9],
       [ 4,  5,  6, 10, 11, 12]])
```

```
r = np.array([1,2,3])
s = np.array([4,5,6])
```

```
np.hstack([r,s])
```

```
array([1, 2, 3, 4, 5, 6])
```

```
np.vstack([r,s])
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
np.column_stack([r,s])
```

```
array([[1, 4],
       [2, 5],
       [3, 6]])
```

```
a
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
a.fill(1)     # change the origin
```

```
a
```

```
array([[1, 1, 1],
       [1, 1, 1]])
```

```
a = np.array([1,2,3,4,5,6])
```

```
a
```

```
array([1, 2, 3, 4, 5, 6])
```

```
a2 = a.copy()
```

```
a2
```

```
array([1, 2, 3, 4, 5, 6])
```

```
a = np.where(a>3, a, 0)
```

```
a
```

```
array([0, 0, 0, 4, 5, 6])
```

```
a2
```

```
array([1, 2, 3, 4, 5, 6])
```

```
b = np.array([1,2,3,4,5])
```

```
b2 = b
```

```
b.fill(1)
```

```
b
```

```
array([1, 1, 1, 1, 1])
```

```
b2
```

```
array([1, 1, 1, 1, 1])
```

```
a = np.array([5,6,1,3,7,8,9,4,2,4])
```

```
print(np.mean(a))
```

```
4.9
```

```
print(np.std(a))
```

```
2.4677925358506134
```

```
print(a.mean())
```

```
4.9
```

```
print(a.std())
```

```
2.4677925358506134
```

```
print(np.argmin(a))
```

```
2
```

```
print(np.argmax(a))
```

```
6
```

```
a = np.array([1,2,3])
```

```
np.repeat(a, 3)
```

```
array([1, 1, 1, 2, 2, 2, 3, 3, 3])
```

```
a = np.array([[1,2,3], [4,5,6], [7,8,9]])
```

```
a
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

```
np.diag(a)
```

```
array([1, 5, 9])
```

```
a > 5
```

```
array([[False, False, False],
       [False, False,  True],
       [ True,  True,  True]])
```

```
a[a > 5]
```

```
array([6, 7, 8, 9])
```

```
a
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

```
print(a[0])
```

```
[1 2 3]
```

```
print(a[0][0])
```

```
1
```

```
a = np.array([4,5,6])
```

```
a
```

```
array([4, 5, 6])
```

```
a[1:]
```

```
array([5, 6])
```

```
a[:] = 2
```

```
a
```

```
array([2, 2, 2])
```

```
a[0] = 99
```

```
a
```

```
array([99,  2,  2])
```

```
a = [1,2,3]
b = [4,5,6]
```

```
a + b
```

```
[1, 2, 3, 4, 5, 6]
```

```
import time
```

```
time.time()
```

```
1742674439.312491
```

```
a = [1,2,3]
b = [4,5,6]
```

```
res = []
for i in range(len(a)):
    res.append(a[i] + b[i])
print(res)
```

```
[5, 7, 9]
```

```
c = np.array([1,2,3])
d = np.array([4,5,6])
```

```
c + d
```

```
array([5, 7, 9])
```

```
type(a)
```

```
list
```

```
type(c)
```

```
numpy.ndarray
```

```
import time
import numpy as np
def nu():
    t1 = time.time()
    a = np.arange(1000000)
    b = np.arange(1000000)
    c = a + b
    t2 = time.time()
    print(t2 - t1)
nu()
```

```
0.021049976348876953
```

```
import time
import numpy as np
def py():
    t1 = time.time()
    c = range(1000000)
    d = range(1000000)
    e = []
    for i in range(len(c)):
        e.append(c[i] + d[i])
    t2 = time.time()
    print(t2 - t1)
py()
```

```
0.5525665283203125
```

numpy random

```
import numpy as np
```

create between [0,1)

```
print(np.random.rand())                # float
print(np.random.rand(5))               # numpy.ndarray
print(np.random.rand(3,3))             # numpy.ndarray
```

```
0.4319982620242404
[0.71994366 0.23677263 0.75543959 0.59827055 0.02094265]
[[0.66756758 0.12333022 0.82816932]
 [0.63266498 0.40782391 0.36952843]
 [0.73692593 0.16554559 0.88943869]]
```

create sample mean: 0 std: 1

```
print(np.random.randn())                # float
print(np.random.randn(5))               # numpy.ndarray
print(np.random.randn(3,3))             # numpy.ndarray
```

```
-0.8788900973284512
[-0.31738978  0.82860969  2.34950818  0.53123701 -1.25522383]
[[-0.24538958 -1.6325537   0.44759907]
 [ 0.64656952  0.97003565  0.50764267]
 [-1.11255323 -0.71368107  0.55674392]]
```

create between [0,1)

```
print(np.random.random())              # float
print(np.random.random(5))             # numpy.ndarray
print(np.random.random([3,3]))         # numpy.ndarray
```

```
0.293264089383504
[0.25159    0.99465444 0.67433425 0.35328299 0.95919577]
[[0.81364272 0.78481079 0.32046827]
 [0.2241887  0.37988658 0.25982006]
 [0.76641305 0.66325929 0.58654919]]
```

create normal sample

```
print(np.random.normal(65, 3))            # float
print(np.random.normal(65, 3, 5))         # numpy.ndarray
print(np.random.normal(65, 3, [3,3]))     # numpy.ndarray
```

```
60.69198667642427
[70.20455498 66.52078038 63.81871932 66.10889321 62.5046141 ]
[[72.47502575 61.11643191 66.62170172]
 [64.87121714 64.31611821 66.25120022]
 [64.7962027  65.37773247 62.65991023]]
```

create random integer between [low, high)

```
print(np.random.randint(0,10))       # int
print(np.random.randint(0,10,5))       # numpy.ndarray
print(np.random.randint(0,10,[3,3]))       # numpy.ndarray
```

```
7
[7 1 9 9 2]
[[4 1 5]
 [6 2 6]
 [4 0 7]]
```

etc

```
a = np.identity(3)
a
```

```
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

```
b = np.arange(15).reshape((3,5))
b
```

```
array([[ 0,  1,  2,  3,  4],
       [ 5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14]])
```

```
np.random.shuffle(b)  # shuffle the rows and do not return
b
```

```
array([[ 5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14],
```

```
       [ 0,  1,  2,  3,  4]])
```

```
c = np.array([ 0,  1,  2,  3,  4])
np.random.choice(c)      # input must be 1D
```

```
np.int64(1)
```

```
a = np.arange(28)
a.reshape((-1, 7))
```

```
array([[ 0,  1,  2,  3,  4,  5,  6],
       [ 7,  8,  9, 10, 11, 12, 13],
       [14, 15, 16, 17, 18, 19, 20],
       [21, 22, 23, 24, 25, 26, 27]])
```